Analysis Description

1. Quantitative Analysis
    a. Sample data came in a CSV (semicolon in fact) file with 7 fields: Id Sequence, Date, Country (which is irrelevant because all the records have “Austria” as Value, Feedback, NPS, and detail in two fields of the feedback split in pros and cons.
    b. We are talking about a sample of 451 answers from January 2023 till March 2024.
    c. Before uploading this file into an Excel Spreadsheet, we massage the information of field feedback because it had on some of the records some other “internal” semicolons that Excel interpreted as a new field, so we deleted them using some replace command on the text file.
    d. Once the info was able to be visualized in Excel properly, we reviewed it and discovered that the ID Sequence field and the pro and con fields could give us some extra insights.
    e. That is why we decided to create two more fields: gap (with a simple formula of subtracting previous minus current) and both (with a logical formula: if both fields are not empty, we put a 1, and if some are empty, we put a 0).
    f. With these two fields we could find the density of answers among all other Bosch operations (the larger the less answer for Austria) and the volume of answers per sentiment.
    g. We produced the three pivot tables with the dates in the rows and in columns:
        i. NPS, Pros, Cons, and Both count
        ii. Average NPS and Average GAP
        iii. NPS per segment from 0 to 10
    h. These three Pivot tables gave us very powerful insight for example we did not have data from March 2023 till June 2023 so there is no information for Q2 2023. And, in this break, NPS went down and took 2 quarters to get up again.
2. Qualitative Analysis
    a. R routine
        i. All the feedback fields were loaded into a Corpus data type variable.
        ii. Using functionality from R, we standardized the words to be all in lowercase with no extra spaces, deleted punctuation marks, and deleted words that we knew would distort the analysis.
        iii. With this bag of words, we discovered the most repetitive ones and created a bar plot.
        iv. We associate the more common (“product”) with the other words next to it.
        v. Then we generated a bigram to understand the more common pair of words.
        vi. All these insights were shown in cloud word diagrams.
        vii. A dendrogram to see pairs of words at levels.
    b. Python code
        i. We used the code provided during class.
        ii. We ran it and produced the two clouds for pros and cons.